

Electricity

MCQ

1. Electricity is defined as the flow of

- A) Protons
- B) Neutrons
- C) Electrons
- D) Atoms

Answer: C

2. Free electrons are electrons that

- A) Are tightly bound to nucleus
- B) Cannot move
- C) Can move from atom to atom
- D) Are inside nucleus

Answer: C

3. Electricity is mainly used for

- A) Only lighting
- B) Only heating
- C) Lighting, heating and operating devices
- D) Decoration

Answer: C

4. An atom consists of

- A) Only electrons
- B) Only protons
- C) Nucleus and electrons
- D) Only neutrons

Answer: C

5. The nucleus of an atom is

- A) Negatively charged
- B) Positively charged
- C) Neutral

D) Variable

Answer: B

6. Electrons revolve around the nucleus in

A) Rings

B) Shells or orbits

C) Lines

D) Layers of protons

Answer: B

7. Electrons in the outermost orbit are

A) Strongly bound

B) Difficult to remove

C) Loosely bound

D) Neutral

Answer: C

8. Electric charge is a

A) Chemical property

B) Physical property

C) Biological property

D) Mechanical property

Answer: B

9. Charges can be

A) Created

B) Destroyed

C) Created and destroyed

D) Neither created nor destroyed

Answer: D

10. Types of electric charges are

A) One

B) Two

C) Three

D) Four

Answer: B

11. Proton carries
- A) Negative charge
 - B) No charge
 - C) Positive charge
 - D) Variable charge

Answer: C

12. Electron carries
- A) Positive charge
 - B) Neutral charge
 - C) Negative charge
 - D) No charge

Answer: C

13. Like charges
- A) Attract
 - B) Repel
 - C) Neutralize
 - D) Destroy

Answer: B

14. Unlike charges
- A) Repel
 - B) Do not interact
 - C) Attract
 - D) Neutralize permanently

Answer: C

15. SI unit of charge is
- A) Ampere
 - B) Volt
 - C) Coulomb
 - D) Ohm

Answer: C

16. Symbol of coulomb is
- A) A

B) V

C) C

D) Ω

Answer: C

17. One coulomb is charge flowing in one second through

A) One volt

B) One ampere

C) One ohm

D) One watt

Answer: B

18. Elementary charge is the charge of

A) Proton only

B) Neutron

C) Electron

D) Atom

Answer: C

19. Elementary charge is denoted by

A) q

B) e

C) c

D) E

Answer: B

20. Atom is electrically neutral because

A) No electrons

B) No protons

C) Equal protons and electrons

D) Only neutrons

Answer: C

21. Transfer of electrons causes

A) Heating

B) Charging

C) Melting

D) Evaporation

Answer: B

22. Substance gaining electrons becomes

A) Positive

B) Neutral

C) Negative

D) Unstable

Answer: C

23. Substance losing electrons becomes

A) Negative

B) Neutral

C) Positive

D) Stable

Answer: C

24. Charging by friction occurs when

A) Two conductors touch

B) Two insulators are rubbed

C) Battery is connected

D) Circuit is closed

Answer: B

25. Charging by conduction requires

A) Rubbing

B) Heating

C) Contact

D) Light

Answer: C

26. Charging by induction occurs

A) With direct contact

B) Without contact

C) Only by friction

D) Only by heating

Answer: B

27. In induction, grounding is used to

- A) Add electrons
- B) Remove electrons
- C) Neutralize charge
- D) Store charge

Answer: B

28. Flow of electrons is from

- A) Positive to negative
- B) Higher to lower potential
- C) Negative to positive
- D) Neutral to charged

Answer: C

29. Conventional current flows from

- A) Negative to positive
- B) Lower to higher potential
- C) Positive to negative
- D) Random direction

Answer: C

30. Potential difference is also called

- A) Current
- B) Resistance
- C) Voltage
- D) Charge

Answer: C

31. Unit of voltage is

- A) Ampere
- B) Volt
- C) Coulomb
- D) Watt

Answer: B

32. Conductors allow

- A) Heat only

- B) Light only
- C) Electric charges to flow
- D) No flow

Answer: C

33. Copper is a

- A) Insulator
- B) Semiconductor
- C) Conductor
- D) Non-metal

Answer: C

34. Rubber is an

- A) Insulator
- B) Conductor
- C) Semiconductor
- D) Electrolyte

Answer: A

35. An electroscope is used to

- A) Measure current
- B) Detect electric charge
- C) Measure voltage
- D) Produce electricity

Answer: B

36. First electroscope was invented by

- A) Newton
- B) Faraday
- C) William Gilbert
- D) Edison

Answer: C

37. Versorium uses a

- A) Gold leaf
- B) Plastic rod
- C) Pivoted needle

D) Wire coil

Answer: C

38. Pith-ball electroscope was invented by

A) Bennet

B) Gilbert

C) Canton

D) Faraday

Answer: C

39. Gold-leaf electroscope was developed by

A) John Canton

B) Abraham Bennet

C) Edison

D) Maxwell

Answer: B

40. Gold is used in electroscope because it is

A) Cheap

B) Insulator

C) Good conductor

D) Magnetic

Answer: C

41. Divergence of gold leaves shows

A) No charge

B) Presence of charge

C) Heating

D) Sound

Answer: B

42. Electrical discharge means

A) Charging

B) Loss of charge

C) Gain of electrons

D) Short circuit

Answer: B

43. Lightning is a

- A) Mechanical process
- B) Chemical reaction
- C) Electrical discharge
- D) Nuclear reaction

Answer: C

44. Thunder is produced due to

- A) Light
- B) Heat expansion of air
- C) Rainfall
- D) Wind

Answer: B

45. Lightning heats air up to

- A) 300°C
- B) 3000°C
- C) 30,000°C
- D) 300,000°C

Answer: C

46. Sound of thunder travels slower than

- A) Light
- B) Heat
- C) Wind
- D) Rain

Answer: A

47. Lightning occurs due to imbalance of

- A) Heat
- B) Charges
- C) Pressure
- D) Wind

Answer: B

48. Earthing transfers excess charge to

- A) Air

- B) Water
- C) Earth
- D) Battery

Answer: C

49. Earthing wire has

- A) High resistance
- B) Medium resistance
- C) Low resistance
- D) Variable resistance

Answer: C

50. Common earthing materials are

- A) Plastic and rubber
- B) Iron and copper
- C) Wood and glass
- D) Aluminium and zinc

Answer: B

51. Lightning arrester protects from

- A) Short circuit
- B) Fire
- C) Lightning stroke
- D) Overloading

Answer: C

52. Electric circuit provides a path for

- A) Heat
- B) Light
- C) Electric current
- D) Sound

Answer: C

53. Source of electricity in a circuit is

- A) Bulb
- B) Switch
- C) Battery

D) Wire

Answer: C

54. A switch is used to

A) Produce electricity

B) Control current

C) Measure current

D) Store charge

Answer: B

55. Resistor converts electrical energy into

A) Light only

B) Heat only

C) Sound

D) Mechanical energy

Answer: B

56. In series circuit current is

A) Different everywhere

B) Same everywhere

C) Zero

D) Random

Answer: B

57. In series circuit voltage is

A) Same across all

B) Zero

C) Divided across components

D) Infinite

Answer: C

58. If one bulb fails in series circuit

A) Others glow brighter

B) Others glow dim

C) All go off

D) No change

Answer: C

59. In parallel circuit voltage across each branch is

- A) Different
- B) Zero
- C) Same
- D) Variable

Answer: C

60. In parallel circuit current

- A) Same in all branches
- B) Divides into branches
- C) Stops
- D) Flows backward

Answer: B

61. Home wiring uses

- A) Series circuit
- B) Parallel circuit
- C) Mixed only
- D) No circuit

Answer: B

62. Effects of current include

- A) Heating only
- B) Chemical only
- C) Heating, chemical and magnetic
- D) Light only

Answer: C

63. Chemical effect occurs in

- A) Metal wire
- B) Conducting solution
- C) Vacuum
- D) Plastic

Answer: B

64. Electrolysis causes

- A) Heating

- B) Decomposition of solution
- C) Expansion
- D) Light emission

Answer: B

65. Electroplating uses

- A) Magnetic effect
- B) Heating effect
- C) Chemical effect
- D) Mechanical effect

Answer: C

66. Heating effect converts electrical energy into

- A) Chemical energy
- B) Mechanical energy
- C) Heat energy
- D) Nuclear energy

Answer: C

67. Wire offering high resistance gets

- A) Cooled
- B) Heated
- C) Neutral
- D) Insulated

Answer: B

68. Tungsten is used in bulbs because it

- A) Has low melting point
- B) Offers high resistance
- C) Is insulator
- D) Is cheap

Answer: B

69. Nichrome is used in heaters because it

- A) Conducts heat poorly
- B) Has high resistance
- C) Melts easily

D) Is magnetic

Answer: B

70. Fuse works on

A) Chemical effect

B) Magnetic effect

C) Heating effect

D) Light effect

Answer: C

71. Fuse wire melts due to

A) Low current

B) High voltage

C) Excess current

D) Low resistance

Answer: C

72. Electric cooker uses

A) Chemical effect

B) Heating effect

C) Magnetic effect

D) Light effect

Answer: B

73. Electric kettle heats water by

A) Radiation

B) Conduction and convection

C) Only radiation

D) Only conduction

Answer: B

74. Electric iron presses clothes using

A) Light energy

B) Sound energy

C) Heat energy

D) Chemical energy

Answer: C

75. Flow of current requires

- A) Open circuit
- B) Closed circuit
- C) Broken wire
- D) Insulator

Answer: B

76. Battery converts

- A) Electrical to chemical
- B) Chemical to electrical
- C) Mechanical to electrical
- D) Heat to electrical

Answer: B

77. Higher potential means

- A) More negative charge
- B) No charge
- C) More positive charge
- D) Equal charge

Answer: C

78. Lower potential means

- A) More electrons
- B) More positive charge
- C) Neutral
- D) No current

Answer: A

79. Electric energy is produced due to

- A) Resting electrons
- B) Moving electrons
- C) Protons
- D) Neutrons

Answer: B

80. Plastic is a

- A) Good conductor

- B) Semiconductor
- C) Insulator
- D) Electrolyte

Answer: C

81. Aluminium is used in wires because it

- A) Is light and conducts electricity
- B) Is insulator
- C) Is non-metal
- D) Is brittle

Answer: A

82. Gold leaf electroscope is more

- A) Simple
- B) Sensitive
- C) Cheap
- D) Heavy

Answer: B

83. Touching electroscope discharges it through

- A) Air
- B) Wire
- C) Earth
- D) Battery

Answer: C

84. Lightning between clouds is called

- A) Ground lightning
- B) Cloud-to-cloud lightning
- C) Earthing
- D) Short circuit

Answer: B

85. Earthing prevents

- A) Electricity generation
- B) Shock
- C) Current flow

D) Voltage

Answer: B

86. GI stands for

A) Ground Insulator

B) Galvanised Iron

C) Gas Insulator

D) General Iron

Answer: B

87. Lightning arrester provides

A) High resistance path

B) Low resistance path

C) Insulated path

D) Broken path

Answer: B

88. Electric bulb glows due to

A) Chemical effect

B) Heating of filament

C) Magnetic effect

D) Electrolysis

Answer: B

89. Current is measured in

A) Volt

B) Ohm

C) Ampere

D) Coulomb

Answer: C

90. Resistance is offered by

A) Path

B) Source

C) Load

D) Switch

Answer: C

91. Filament glows because it is

- A) Thick
- B) Thin with high resistance
- C) Insulator
- D) Magnetic

Answer: B

92. Parallel circuit prevents

- A) Voltage division
- B) Current flow
- C) Bulb failure
- D) Energy use

Answer: A

93. Series circuit is used in

- A) Home wiring
- B) Street lights rarely
- C) Decoration lights
- D) Motors

Answer: C

94. Electric current is flow of

- A) Protons
- B) Neutrons
- C) Electrons
- D) Atoms

Answer: C

95. Heating effect increases with

- A) Lower resistance
- B) Higher resistance
- C) No resistance
- D) Zero current

Answer: B

96. Chemical effect helps in

- A) Cooking

- B) Lighting
- C) Metal purification
- D) Sound production

Answer: C

97. Electroplating deposits

- A) Plastic
- B) Gas
- C) Metal
- D) Salt

Answer: C

98. A blown fuse indicates

- A) Normal current
- B) Excess current
- C) Low voltage
- D) No resistance

Answer: B

99. Thunder is heard after lightning because

- A) Sound travels faster
- B) Light travels faster
- C) Air absorbs sound
- D) Wind blocks sound

Answer: B

100. Electricity is a form of

- A) Matter
- B) Energy
- C) Force
- D) Element

Answer: B